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Enhancing IIoT security: a hybrid AI framework for intrusion detection powered by Gen-X AI

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ABSTRACT

Intrusion detection in Industrial Internet of Things (IIoT) systems is critical for protecting essential infrastructure from cyber threats. This paper introduces a novel multistage optimization framework designed to enhance intrusion detection accuracy. The first stage employs a Hybrid ACO-GA/PSO algorithm for feature selection, ensuring the retention of the most relevant features to improve classification precision. In the second stage, Long Short Term Memory (LSTM) networks are optimized using a combination of the Bat Algorithm and the Whale Optimization Algorithm, enabling precise fine-tuning of hyper-parameters to maximize model performance. A Generative Adversarial Network (GAN) is incorporated in the final evaluation stage, testing the model's robustness against previously unseen data, followed by training the generated data on the optimized LSTM model. To further interpret and explain the results, we apply Explainable AI (XAI) techniques, including SHAP and LIME, providing insight into feature importance and model decision-making. Our approach is evaluated with standard performance metrics, demonstrating significant improvements in detection accuracy and validating the efficacy of our hybrid framework for advancing IIoT security mechanisms.

Highlights

- Multi-Stage Optimization Framework
- Hybrid Feature Selection
- Fine-tuned LSTM Model
- GAN for Robustness Testing
- Explainable AI (XAI) Integration

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Industrial internet of things; intrusion detection; hybrid feature selection; generative artificial intelligence; explainable artificial intelligence

1. Introduction

The evolution of technology has been marked by significant milestones, with the advent of the Internet being one of the most transformative. The Internet has revolutionized communication, commerce and information sharing, leading to the development of the Internet of Things (IoT), a network of interconnected devices capable of collecting and exchanging data. IoT has found applications in various sectors, including healthcare, transportation, and home automation, enhancing efficiency and convenience [1]. The Industrial Internet of Things (IIoT) extends the concept of IoT to industrial settings, integrating sensors, machines, and analytics to optimize industrial processes. IIoT enables real-time monitoring, predictive maintenance, and improved operational

efficiency, but it also introduces new security challenges that must be addressed to protect critical infrastructure [2].

The rapid growth of the Industrial Internet of Things (IIoT) has revolutionized industrial operations, enabling enhanced efficiency, productivity, and automation. However, this increased connectivity has also exposed industrial systems to a multitude of security threats. Traditional security measures, such as specialized operating systems and hardening procedures, have proven insufficient against evolving attack vectors and the discovery of new vulnerabilities [3].

Attack vectors in IIoT environments span a wide range of techniques used by adversaries to exploit system weaknesses. These include unauthorized access to IIoT devices

and networks, denial-of-service (DoS) attacks that disrupt operations, and malicious command injection targeting control systems. Common threats also involve reconnaissance attacks to map the network, man-in-the-middle (MitM) attacks to intercept or alter communications, and protocol exploits targeting industrial communication standards such as Modbus and MQTT. Attackers may also use lateral movement to access more critical systems, perform IP spoofing and in-path attacks, or exploit third-party access through vendors and contractors. Additionally, IIoT systems are vulnerable to phishing attacks, credential theft, data integrity breaches, and ransomware, which pose serious risks to industrial safety and reliability. Real-world incidents like the Stuxnet worm, TRITON malware, and the Colonial Pipeline ransomware attack underscore the tangible consequences of these threats, including physical damage, operational shutdowns, and economic disruption [4–7].

Intrusion detection has emerged as a critical security measure to safeguard IIoT networks from malicious attacks. Conventional intrusion detection systems face inherent limitations in addressing the unique challenges posed by the large-scale, distributed, and resource-constrained nature of IIoT environments [8]. To address these challenges, researchers have turned to advanced artificial intelligence techniques, such as fuzzy logic, machine learning, and Bayesian networks, which have shown promising results in detecting malicious behaviors and anomalies in IIoT devices [9, 10].

In this paper, we propose a comprehensive, multi-stage intrusion detection framework designed for IIoT systems. In the first stage, we employ a hybrid feature selection method combining Ant Colony Optimization (ACO), Genetic Algorithm (GA), and Particle Swarm Optimization (PSO) to identify the most relevant attributes for intrusion detection. This hybrid optimization enhances feature selection, improving classification accuracy and reducing computational complexity. In the second stage, the Bat Algorithm and Whale Optimization Algorithm are used to fine-tune the hyper-parameters of a Long Short-Term Memory (LSTM) model, optimizing its performance in accurately detecting and classifying network traffic as normal or malicious.

A critical addition in our framework is the incorporation of Generative Adversarial Networks (GAN) in an evaluation stage, where we test the robustness of the intrusion detection model against synthetic, previously unseen data. This helps assess the model's generalization capability to new attack scenarios. Furthermore, the GAN generated data is trained on the optimized LSTM model to enhance the model's resilience to diverse threats. To provide insight into the model's decision-making process, we apply Explainable AI (XAI)

techniques, specifically SHAP and LIME, on the model trained with GAN generated data. These techniques allow us to analyze feature importance and explain the model's predictions, ensuring transparency and trust in the intrusion detection process.

We evaluate our approach using the X-IIoTID dataset, a comprehensive benchmark for IIoT intrusion detection. By integrating multiple optimization techniques and leveraging GANs for further evaluation and training, our approach aims to overcome limitations such as low precision, high complexity, and scaling difficulties present in traditional intrusion detection systems. Through this hybrid AI approach, we strive to develop a robust, efficient, and interpretable intrusion detection solution capable of effectively identifying and mitigating security threats in IIoT environments.

This research advances IIoT security by demonstrating the potential of integrating advanced optimization and generative techniques alongside explainable models to enhance intrusion detection mechanisms and improve the overall security posture of IIoT systems.

2. Related works

The increasing complexity and volume of data generated in IIoT networks necessitate advanced methods for network intrusion detection and anomaly detection. Traditional approaches often struggle with high-dimensional data and dynamic attack patterns, highlighting the need for more sophisticated techniques. Recent studies have explored the use of many techniques including deep learning, hybrid meta-heuristics, optimization algorithms, generative AI and explainable ai to address these challenges.

2.1. Deep learning-based intrusion detection systems

Deep learning (DL) techniques have been extensively employed to enhance intrusion detection in IIoT environments. For instance, Chaurasia et al. [11] developed a ResNet-based Intrusion Detection System (IDS) utilizing federated learning, achieving impressive accuracy rates of 99.43% in centralized and 99.16% in federated settings on the X-IIoTID dataset. In a complementary line of work, Jianping et al. [12] leverage federated learning with attention-based Graph Neural Networks (GNNs) to detect network attacks while preserving data privacy. This method not only enables collaborative learning across distributed IIoT devices but also improves detection of complex graph-structured network behaviors through attention mechanisms. Their results show strong performance without compromising

sensitive edge data, aligning with emerging needs in decentralized IIoT environments. Similarly, Wang et al. [13] introduced ET-DCANET, a hierarchical IDS that combines random tree feature selection with dual attention mechanisms, attaining high accuracy across multiple datasets, 99.68%, 98.50%, and 99.85% accuracy on NSL-KDD, UNSW-NB15, and X-IIoTID datasets, respectively. In Addition, Qathrady et al. [14] introduced SACNN, a Self-Attention Convolutional Neural Network that uses self-attention for feature extraction and CNNs for detection. Their system was validated using the Edge-IIoTset and X-IIoTID datasets, covering a wide range of IIoT communication protocols and attack scenarios. To enhance generalizability across datasets, Babayigit et al. [15] proposed a unified DL framework using a hybrid CNN-GRU model and manifold alignment through autoencoders and locally linear embedding. Their model, evaluated on Edge-IIoTset, WUSTL-IIoT-2021, and X-IIoTID datasets, achieved 97.68% accuracy and demonstrated improved transfer learning performance, increasing accuracy from 50.95% to 97.80%. Moreover, Alshehri et al. [16] developed SA-DCNN, incorporating self-attention mechanisms for feature importance and deep CNNs for classification. Their approach used mutual information-based feature filtering and a two-step data cleaning process, achieving superior performance on IoTID20 and Edge-IIoTset datasets.

While these architectures underscore the effectiveness of DL in IIoT intrusion detection, most rely on static hyperparameter configurations or single-stage optimization. In contrast, our work introduces a multi-stage DL pipeline that integrates metaheuristic optimization for hyperparameter tuning and leverages generative models for robust evaluation, thus addressing performance, adaptability, and explainability in a unified framework.

2.2. Optimization-driven feature selection and performance enhancement in IDS

Optimization algorithms have become essential in improving the performance of intrusion detection systems (IDS), particularly in automating feature selection and fine-tuning model parameters to handle high-dimensional and imbalanced data common in IIoT environments. For example, Abdulkareem et al. [17] proposed a feature selection and a lightweight ensemble learning approach that uses feature importance (FI) for dimensionality reduction. The model was applied to the Edge-IIoTset dataset, which includes IoT and IIoT network records, and it achieved an accuracy of 87.37%. Likewise, Lu et al. [18] addressed class imbalance, redundant features, and inadequate detection accuracy in IIoT intrusion detection using hierarchical clustering with

under-sampling to balance class distributions and a greedy approach for optimal feature selection. The proposed deep neural network model with parallel global and local sub-networks significantly enhanced detection performance.

While these approaches effectively demonstrate the potential of heuristic and deterministic optimization methods, they often rely on static or single-stage optimization pipelines. Our work builds upon this by introducing a hybrid multi-stage optimization framework that combines metaheuristic algorithms such as ACO, GA, PSO, Bat, and Whale Optimization for both feature selection and LSTM hyperparameter tuning. This comprehensive optimization strategy not only enhances model accuracy but also improves its robustness in complex IIoT environments.

2.3. Generative AI-based approaches

The rise of generative AI, particularly Generative Adversarial Networks (GANs), has opened new avenues in addressing long-standing challenges in intrusion detection systems (IDS), such as data imbalance and poor generalization to novel attack types. Constantin et al. [19] focuses on leveraging Generative Adversarial Networks (GANs) to address the challenge of imbalanced network traffic datasets, which often contain significantly more ‘normal’ traffic than ‘intrusion’ samples. By evaluating various GAN architectures such as Wasserstein, conditional, and LSTM GANs the authors generate synthetic intrusion data to augment flow-based network traffic datasets. The study demonstrates how this augmentation improves the performance of neural networks in intrusion detection tasks. Rahman et al. [20], address the challenge of detecting diverse attack classes in imbalanced datasets, like NSL-KDD, which often lead to biased models favoring dominant classes. To overcome this, they employ a Conditional Tabular Generative Adversarial Network (CTGAN) to generate synthetic samples, effectively balancing the dataset. They then apply a deep neural network with a 4-neuron output layer for multi-class classification.

While these approaches demonstrate improved detection through data augmentation, they primarily focus on balancing existing datasets rather than evaluating model robustness on entirely novel data distributions. Additionally, limited attention is given to interpreting how synthetic data influences model decisions, underscoring the need for combining generative approaches with explainable AI techniques.

In contrast to these efforts, our work integrates GAN-based data generation with optimized LSTM models and explainable AI, aiming not only to enhance robustness

against unseen attacks but also to interpret the model's behavior through SHAP and LIME.

2.4. Explainable AI (XAI) in intrusion detection

As deep learning models gain popularity in intrusion detection systems (IDS), especially in mission-critical environments like the Industrial Internet of Things (IIoT), the need for transparency and interpretability has become paramount. Explainable AI (XAI) techniques are increasingly used to bridge the gap between model accuracy and decision interpretability. Attique et al. [21] introduced an explainable and intelligent IDS combining BiLSTM with a self-adaptive attention mechanism that dynamically prioritizes critical data segments. The system achieved high performance, with accuracies of 99.92% on CICIDS2017 and 96.54% on X-IIoTID, and enhanced transparency with SHapley Additive explanations (SHAP) from Explainable AI. Sharma et al. [22] proposed two Deep Learning models for IoT intrusion detection, focusing on feature selection and attack classification using the NSL-KDD and UNSW-NB 15, and in order to enhance the interpretability of these models, they applied xAI techniques like SHAP and LIME for offering greater transparency and trust in the DL model's decisions. Arreche et al. [23] introduces a framework to evaluate black-box explainable AI (XAI) methods for network intrusion detection systems (IDS). It assesses two popular techniques, SHAP and LIME, using six metrics: descriptive accuracy, sparsity, stability, efficiency, robustness, and completeness. Tested on three intrusion datasets and seven AI models, the framework highlights both strengths and limitations of XAI methods in IDS, offering a valuable tool for security analysts to better interpret AI-driven intrusion detection.

While existing studies demonstrate the utility of SHAP and LIME in visualizing and interpreting DL-based IDS decisions, they often lack integration with advanced data generation or augmentation methods, such as GANs. Moreover, most works provide a limited empirical assessment of explanation quality or robustness in real-time IIoT scenarios. Our work addresses these limitations by incorporating XAI in conjunction with generative learning, offering both robust performance and meaningful insight into the decision-making process of intrusion detection models.

2.5. Limitations and challenges

While recent studies have made significant strides in advancing intrusion detection for IIoT systems, several critical challenges remain. Many works, such as those employing hierarchical attention mechanisms or

ensemble learning techniques, achieve high accuracy on specific datasets like X-IIoTID or NSL-KDD [13, 17]. However, their generalizability across diverse IIoT datasets with varying attack types is often unverified, raising concerns about deployment in real-world heterogeneous environments.

Additionally, although hybrid meta-heuristic approaches have been proposed for feature selection and hyperparameter tuning, they typically focus on performance improvement without thoroughly evaluating robustness under unseen conditions or adversarial scenarios. Most of these methods also use single-stage or static optimization, which limits adaptability in dynamic IIoT contexts.

Similarly, applications of Generative Adversarial Networks (GANs) in intrusion detection primarily aim to augment datasets to alleviate class imbalance [19, 20]. These efforts often neglect using GANs for robustness testing under novel, synthetic threat landscapes, a gap that limits their utility in real-world, evolving attack environments. Moreover, little emphasis has been placed on interpreting how GAN-generated data influences decision-making, limiting their transparency and auditability.

Furthermore, while Explainable AI (XAI) techniques like SHAP and LIME have been employed to visualize and understand model behavior [22, 23], their integration with complex, hybrid learning systems remains limited. Few studies provide a detailed analysis of explanation robustness, computational overhead, or real-time applicability in resource-constrained IIoT systems.

These limitations underscore the need for a holistic and scalable framework that not only integrates advanced optimization and deep learning techniques, but also rigorously evaluates robustness using generative models and improves interpretability with XAI. Our proposed approach directly addresses these gaps by combining hybrid multi-stage optimization, generative evaluation, and explainable analysis in a unified pipeline tailored for IIoT intrusion detection.

3. Methodology

Our research aims to develop a robust and efficient machine learning framework for IIoT intrusion detection using a hybrid optimization approach aided with generative and explainable AI. Specifically, we enhance feature selection by leveraging a hybrid Ant Colony Optimization, Genetic Algorithm, and Particle Swarm Optimization (H-AGP) approach. This hybrid method identifies the most relevant features from the X-IIoTID dataset, improving both the model's accuracy and efficiency.

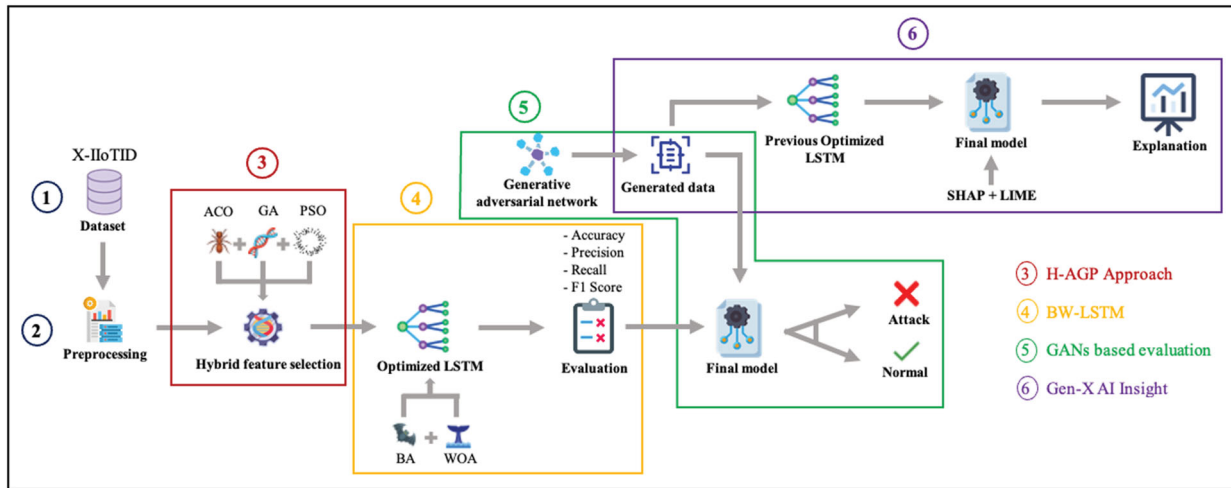


Figure 1. Our proposed methodology.

After feature selection, we optimize the Long Short-Term Memory (LSTM) model by applying the Bat Algorithm (BA) and Whale Optimization Algorithm (WOA) to fine-tune its network parameters, ensuring the model performs optimally in detecting intrusions.

Our approach focuses on accurately classifying network traffic into normal or attack categories, enhancing the model's ability to distinguish between benign and malicious activities. To validate the robustness of the optimized LSTM model, we incorporate Generative Adversarial Networks (GAN) in an evaluation stage, where synthetic, previously unseen data is used to assess the model's generalization capabilities in detecting new attacks.

Furthermore, we implement Explainable AI (XAI) techniques specifically SHAP and LIME on the model trained with the GAN generated data. This allows us to interpret the model's predictions by analyzing feature importance, ensuring transparency and enhancing trust in the model's decision-making process. Figure 1 briefly explains our contribution.

3.1. Dataset collection and preprocessing

In this research, we implemented our contribution using the X-IIoTID dataset to arrive at a powerful approach for intrusion detection in the industrial Internet of Things. It encompasses a variety of network traffic and event logs related to IIoT devices, offering a comprehensive view of the operational and security aspects of IIoT environments. The dataset includes 68 features we will use to detect malicious traffic in IIoT systems and enhance security. Before starting our process, we prepared our data to be cleaned and ready for applying our methodology. A dataset preparation can be made by several steps, including:

- *Removing irrelevant features*
- *Handling missing values*: is a process where we identify and substitute missing data points in a dataset with appropriate values, such as the mean or mode of each feature, to enhance the performance and reliability of predictive models [24].
- *Data balancing*: Techniques to address class imbalance in datasets, such as oversampling, under sampling, and generating synthetic data, to ensure unbiased and accurate model performance [25].
- *Removing high correlated features*: The process of identifying and eliminating redundant features that are highly correlated with each other to prevent multicollinearity, improve model interpretability, and enhance predictive performance [26].
- *Rescaling and normalization*: Rescaling and normalization is a process where we standardize feature values in a dataset to a uniform [0–1] range [27].

3.2. Feature selection using H-AGP approach

Feature selection is a crucial step in developing a predictive model, particularly in fields where precision is paramount. An optimal feature set improves the model's accuracy and boosts computational efficiency by removing irrelevant or redundant attributes, which helps prevent overfitting and reduces processing time [28]. To enhance feature selection in IIoT intrusion detection, we propose the H-AGP Approach, a hybrid method combining the strengths of Ant Colony Optimization (ACO), Genetic Algorithm (GA), and Particle Swarm Optimization (PSO) to create a powerful feature selection technique.

Ant Colony Optimization (ACO) is an algorithm inspired by ant foraging behavior, where artificial ants cooperate to find optimal solutions. Ants lay pheromones

that influence path selection, and over time, paths with higher pheromone levels are more likely to be chosen. This positive feedback mechanism and parallel exploration of solutions make ACO effective and robust, allowing it to escape local optima [29].

Genetic Algorithms (GA) are another class of nature-inspired optimization techniques based on the principles of natural selection and genetics. GAs operate on a population of potential solutions, applying genetic operators such as selection, crossover, and mutation to evolve solutions over generations [30].

Particle Swarm Optimization (PSO) is an algorithm inspired by the social behavior of birds or fish. A swarm of particles, each representing a potential solution, moves through the solution space by adjusting positions based on their own experience and the best solutions found by neighboring particles. PSO involves sharing information among particles to guide them toward optimal regions and is known for its simplicity and minimal parameter adjustments [31].

3.2.1. H-AGP approach

Our proposed leverages the positive feedback mechanism of ACO to probabilistically select features, the evolutionary power of GA to generate new feature subsets through crossover, and the social behavior of PSO to iteratively refine feature subsets based on velocities influenced by personal and global best positions. Together, these methods effectively explore and exploit the solution space to identify optimal feature subsets. The parameters for ACO, GA, and PSO were set based on values commonly used in related literature, ensuring a balanced trade-off between exploration and exploitation. No dedicated parameter tuning or meta-optimization process was applied in this study. The process of our proposed approach passes by the following steps:

- (1) *Initialize parameters*: Start by setting up parameters for the ACO, GA, and PSO algorithms, including pheromone levels and the initial population.
- (2) *Initial fitness calculation*: Compute the fitness of the initial population using the Random Forest algorithm.
- (3) *ACO Phase*: Each ant constructs a solution based on pheromone trails, and the fitness of each solution is evaluated. Pheromone levels are updated based on the fitness of the solutions.
- (4) *GA Phase*: Select the best individuals from the ACO solutions. Perform crossover operations to produce offspring, ensuring that each offspring has at least one feature selected. Compute the fitness of the offspring and replace the least fit individual in the population with the new offspring.

- (5) *PSO Phase*: Initialize particle positions and velocities using the best solutions from the ACO and GA phases. Update the velocities and positions of particles, evaluate their fitness, and update personal and global best positions based on fitness.
- (6) *Best individual selection*: After completing the iterations, select the best individual from the final population.

The Algorithm 1 outlines the steps of the proposed H-AGP approach, which combines ACO, GA, and PSO for optimal feature selection.

This hybrid approach offers numerous advantages. ACO drives exploration through pheromone-guided search, GA introduces genetic variation, and PSO refines solutions with social information sharing. This combination helps circumvent local optima and broaden the solution space. The synergistic effects of each algorithm address individual limitations: GA's genetic diversity, PSO's fast convergence, and ACO's path reinforcement. Moreover, this hybrid approach is flexible and adaptive, adjusting its balance of exploration and exploitation according to the problem's characteristics. Kiran et al. [32], Goldberg [30].

3.3. Data training using BW-LSTM approach

The BW-LSTM Approach integrates the Bat Algorithm and Whale Optimization Algorithm to fine-tune hyperparameters of LSTM networks. This hybrid optimization method aims to enhance model performance and accuracy.

Long Short-Term Memory (LSTM) networks are a type of recurrent neural network (RNN) specifically designed to overcome the limitations of traditional RNNs, particularly the vanishing and exploding gradient problems. LSTMs achieve this through a complex cell structure that allows them to maintain and update information over long periods [33].

The Bat Algorithm, developed by Xin-She Yang in 2010, is inspired by the echolocation behavior of bats. Bats emit sound pulses and use the echoes to locate objects and prey. Key features of the algorithm include echolocation for navigating the search space, frequency tuning for adjusting exploration and exploitation, and dynamic changes in loudness and pulse emission to balance local and global search [34].

The Whale Optimization Algorithm (WOA), introduced by Seyedali Mirjalili and Andrew Lewis in 2016, is inspired by the hunting behavior of humpback whales, particularly the bubble-net strategy. Key elements include encircling prey to update positions, using spiral movements to mimic bubble-net attacks, and alternating

Algorithm 1 H-AGP Approach

Input: Set parameters: number of individuals (P), number of generations (m), number of ants, number of particles, pheromone levels, evaporation rate, encode each individual in the binary vector.

Output: Best individual determined $P(n)$

```

1: Initialize parameters for ACO, GA, and PSO
2: Initialize pheromone levels for ACO
3: Produce initial population for GA/PSO
4: for all individual  $j$  in  $P$  do
5:   Calculate fitness  $J_j$  using the Random Forest algorithm
6: end for
7: while iteration number  $< m$  do                                     ▷ ACO Phase
8:   for all ant  $k$  in the ACO colony do
9:     Construct a solution  $S_k$  based on pheromone trails
10:    Evaluate fitness  $F(S_k)$  using the fitness function
11:  end for
12:  Update pheromone levels based on solutions  $S_k$                                ▷ GA Phase
13:  Select the best individuals  $B$  from ACO solutions
14:  for all pair of parents selected from  $B$  do
15:    Perform crossover to produce offspring
16:    Ensure at least one feature is selected in the offspring
17:  end for
18:  Compute evaluation fitness of offspring
19:  Replace the least fit individual with new offspring                               ▷ PSO Phase
20:  Initialize particle positions and velocities with best solutions from ACO/GA
21:  for all particle  $p$  in the population do
22:    Update velocities and position of particles
23:    Evaluate fitness  $F(p)$ 
24:    Update personal and global best positions based on fitness
25:  end for
26:  Select the final solution based on highest fitness value
27:  Increment iteration number
28: end while
29: Select the best individual  $P(n)$ 

```

between shrinking encircling and spiral updating to balance exploration and exploitation [35].

3.3.1. BW-LSTM approach

The combined Bat Algorithm and Whale Optimization Algorithm (WOA) approach aims to optimize the hyper-parameters and weights of LSTM networks. This hybrid method leverages the echolocation-based search capabilities of the Bat Algorithm and the social hunting strategy of WOA to enhance the performance and efficiency of LSTM models. The process of fine-tuning the LSTM hyper-parameters is present in the following steps:

(1) *Initialization:* Parameters for BA and WOA are initialized, and populations of bats and whales are generated randomly. Each individual represents a potential LSTM hyperparameter set.

(2) *Fitness evaluation:* The initial fitness of bats and whales is evaluated using an LSTM-based fitness function.

(3) *BA Phase:*

- Each bat updates its frequency, velocity, and position.
- New solutions are generated and evaluated. Better solutions update the bat's position and fitness.
- The global best bat is updated if a new best solution is found.

(4) *WOA Phase:*

- Each whale updates its position using encircling, bubble-net, or random search strategies.
- New solutions are evaluated and better solutions update the whale's position and fitness.
- The global best whale is updated if a new best solution is found.

Algorithm 2 BW-LSTM**Input:** Number of bats n_{bats} , number of whales n_{whales} , number of iterations $n_{iterations}$ **Output:** Best LSTM hyper-parameters

```

1: Initialize parameters for BA and WOA
2: Initialize bats and whales populations randomly
3: Evaluate initial fitness of bats and whales using LSTM-based fitness function
4: while iteration number <  $n_{iterations}$  do ▷ BA Phase
5:   for all bat  $i$  do
6:     Update frequency, velocity, and position of bat  $i$ 
7:     Apply simple bounds/constraints to bat  $i$ 
8:     if random number > pulse rate then
9:       Generate a new solution around the best solution
10:    end if
11:    Evaluate new fitness using LSTM model
12:    if new solution is better and random number < loudness then
13:      Update position and fitness of bat  $i$ 
14:    end if
15:  end for
16:  Update global best bat if needed ▷ WOA Phase
17:  for all whale  $i$  do
18:    Update position of whale  $i$  based on encircling, bubble-net, or random search
19:    Apply simple bounds/constraints to whale  $i$ 
20:    Evaluate new fitness using LSTM model
21:    if new solution is better then
22:      Update position and fitness of whale  $i$ 
23:    end if
24:  end for
25:  Update global best whale if needed
26:  Compare the best solution from BA and WOA
27:  Select the best solution as global best solution
28:  Increment iteration number
29: end while
30: Select the best LSTM hyper-parameters

```

- (5) **Comparison and selection:** The best solutions from BA and WOA are compared, and the overall best solution is selected.
- (6) **Iteration increment:** The loop continues until the maximum number of iterations is reached.
- (7) **Final selection:** The best LSTM hyper-parameters are chosen based on the global best solution

The Algorithm 2 outlines the steps of the proposed BW-LSTM approach, which combines BA and WOA for an optimal LSTM model.

Combining the Bat Algorithm (BA) and Whale Optimization Algorithm (WOA) to optimize LSTM networks brings several key benefits. BA enhances exploration through its frequency adjustment and loudness modulation, while WOA focuses on exploitation with its encircling and bubble-net strategies. This hybrid method effectively avoids local optima by employing diverse

search tactics and exploring a broader solution space. The interplay between BA's capacity for genetic diversity and WOA's rapid convergence strengthens successful solution paths. Additionally, this approach is adaptable, dynamically balancing exploration and exploitation to meet varying problem landscapes [34, 35].

3.4. Evaluation and transparency

To assess the performance of the classification system, evaluation metrics were used, including accuracy, which is the percentage of the correct predictions [36], it calculated using the Equation (1).

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

Precision which is the measure that indicates how many of the predicted positive instances are actually correct

[37], it calculated using the Equation (2).

$$\text{Precision} = \frac{TP}{TP + FP} \quad (2)$$

Recall which is the proportion of positive samples that have been accurately classified as positive [38], it calculated using the Equation (3).

$$\text{Recall} = \frac{TP}{TP + FN} \quad (3)$$

The F1-score is the recall and precision harmonic means which strikes a harmony between the two measures, [39] it calculated using the Equation (4).

$$\text{F1-Score} = \frac{2 * \text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4)$$

To further enhance the robustness of our intrusion detection model and ensure its ability to generalize to new, unseen attacks, we incorporated Generative Adversarial Networks (GAN) into the evaluation phase. Generative Adversarial Networks (GANs) are a category of artificial intelligence algorithms that consist of two competing neural networks: the generator and the discriminator. These networks are trained simultaneously in an adversarial process. The generator's role is to create realistic synthetic data, while the discriminator learns to distinguish between real and generated data. Through repeated training cycles, GANs can produce high-quality synthetic data that closely mimics the distribution of real data [40, 41]. In this research, we contribute by using GANs to generate synthetic data similar to the existing data in our dataset, and use it to evaluate our model and ensuring its ability to generalize effectively to unseen data.

Explainable AI (XAI) techniques, specifically SHapley Additive exPlanations (SHAP) and Local Interpretable Model-agnostic Explanations (LIME), are applied to the model trained on GAN generated data to ensure transparency and interpretability. XAI helps make complex machine learning models more understandable by highlighting how input features influence predictions [42]. SHAP provides a global perspective by assigning importance values to each feature, while LIME offers local interpretability by approximating individual predictions [43]. We focus on using XAI with the GAN generated data because this stage introduces novel, synthetic attack patterns that the model has not encountered during training, making it crucial to understand the model's behavior under these new conditions. Since the optimization phase using the Bat Algorithm (BA) and Whale Optimization Algorithm (WOA) aims to fine-tune the LSTM for performance, we concentrate on using XAI where it adds the most value in evaluating the model's generalization on unseen data generated by GANs.

Table 1. Final selected features and their descriptions.

Feature	Description
duration	Duration of connection
des_bytes	Bytes sent to the destination
src_bytes	Bytes sent from the source
is_syn_only	Indicator if the packet is SYN only
read_write_physical.process	Number of read/write operations in the physical process

Table 2. Hyper-parameters of H-AGP approach.

Parameter	Value
Number of Individuals	30
Number of Generations	30
Number of Ants	16
Number of Particles	20
Evaporation Rate	0.5
Influence of pheromone (Alpha)	1.0
Influence of heuristic information	2.0
Fitness function	Random Forest

4. Experiments and results

This section presents the experimental part and the results obtained from applying our proposed approaches to the task of making an intrusion detection system in Industrial Internet of Things (IIoT) using the X-IIoTID dataset.

4.1. Data preprocessing and feature selection

The initial dataset was preprocessed by removing irrelevant features like source and destination IP address (Des_IP and Src_IP) and handling missing values. After this process, we got 59 features. Then, we removed the highly correlated features with a threshold of 0.7. This last step resulted in a dataset with 32 features, which was then used for feature selection and model training. The Figure 2 illustrates the features heatmap before and after removing the correlation.

For feature selection, we utilized our H-AGP approach, which combines the strengths of Ant Colony Optimization (ACO), Genetic Algorithm (GA), and Particle Swarm Optimization (PSO). This hybrid approach resulted in the selection of five features shown in the Table 1, with an accuracy of 99.5%.

Our H-AGP approach effectively combines ACO, GA, and PSO, leading to highly accurate feature selection based on the parameters shown in Table 2

Our approach also provided a comparable results against the ACO/GA approach and the ACO/PSO approach. These results are shown in Table 3.

The Figure 3 illustrates the progression of the fitness value over multiple generations during the optimization process using the H-AGP approach. The *x*-axis represents the number of generations, while the *y*-axis

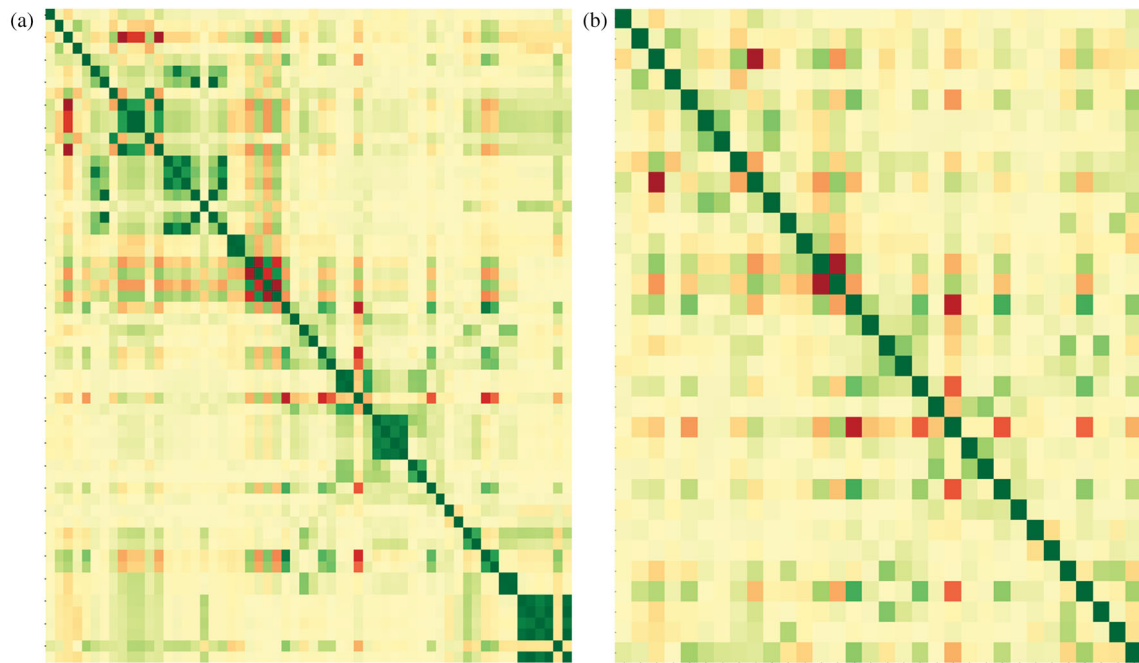


Figure 2. Features correlation heatmap. (a) Heatmap before removing correlation. (b) Heatmap after removing correlation.

Table 3. Comparative results of our proposed approach, ACO/GA, and ACO/PSO.

Approach	Accuracy
ACO/GA	0.990
ACO/PSO	0.992
H-AGP	0.995

indicates the fitness value. The trend shows how the algorithm converges towards an optimal solution, with the fitness value improving over time, demonstrating the effectiveness of our hybrid optimization method in enhancing model accuracy.

4.2. Model training using BW-LSTM approach

After selecting the most relevant features, we employed our hybrid optimization approach which combining the Bat Algorithm and Whale Optimization Algorithm (BW-LSTM), to optimize the parameters of the Long Short-Term Memory (LSTM) network for intrusion detection systems in Industrial Internet of Things (IIoT) systems.

The Table 4 shows the initialized parameters for the process of the BW-LSTM approach and Table 5 define the search space for the hyper-parameters.

After the optimization process, we achieved an optimal LSTM model with an accuracy of 99.94% for the intrusion detection task. To assess the performance of our optimized LSTM, we compared it to other commonly used machine learning and deep learning techniques, specifically Deep Neural Networks (DNN) and

Table 4. Initialized parameters of BW-LSTM approach.

Parameter	Value
Number of Bats	20
Number of Whales	20
Number of Ants	16
Number of Particles	20
Evaporation Rate	0.5
Frequency bounds	Min = 0; Max = 2
Loudness	0.5
Pulse rate	0.5
Alpha	0.9
Gamma	0.9
Number of Iterations	30

Table 5. Search space for the hyper-parameters.

Hyper-parameters	Hyperparameter values/search space
Activation function	ReLU, Tanh, Sigmoid
Units	32, 64, 128, 512
Dropout	[0.0–0.99]
Learning rate	[$1e^{-5}$ –1]
Epoch	8, 12, 32, 64

Table 6. Comparison of our approach with other AI techniques.

Approach	Accuracy	Training Time (s)	Inference Time (ms/sample)
Decision Tree	0.9800	5	1
Random Forest	0.9910	12	5
Deep Neural Network	0.9970	200	12
Standard LSTM	0.9802	250	16
BW-LSTM	0.9994	300	18

Random Forest, Decision Tree (DT) and standard LSTM (with default parameters). The results of these models are presented in Table 6.

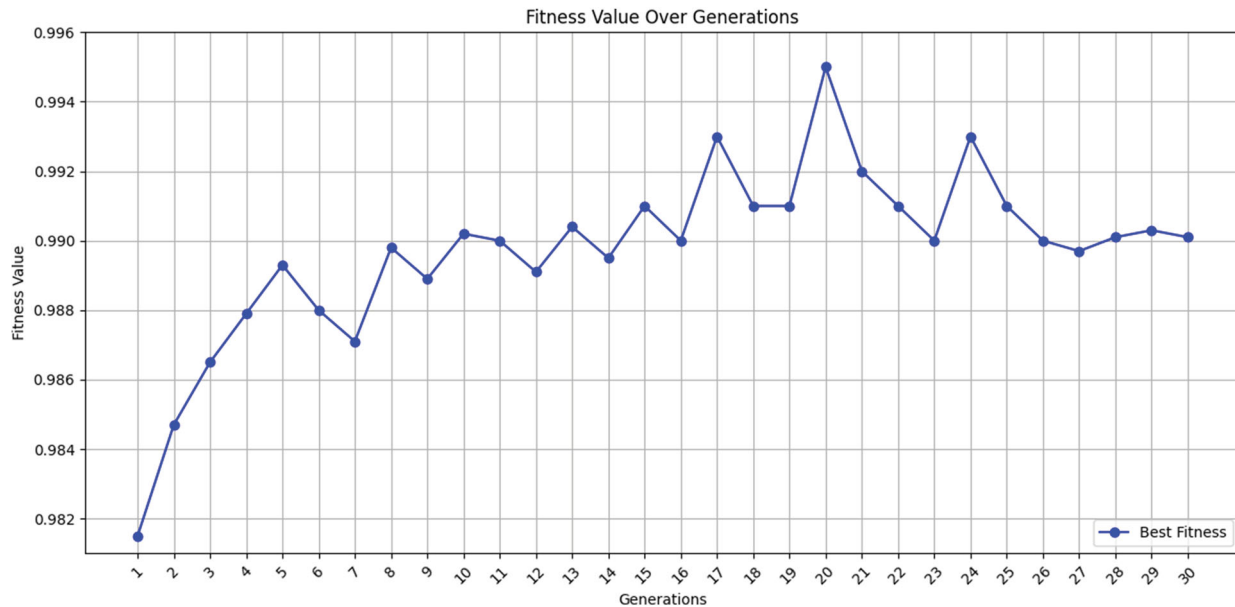


Figure 3. Progression of the fitness value during the optimization process.

Table 7. Evaluation results.

Evaluation metric	Value
Precision	0.9898
Recall	0.9990
F1-Score	0.9994
Accuracy	0.9994

The performance of the optimized LSTM model was evaluated using the most well-known classification metrics. These metrics and their corresponding values are presented in Table 7.

4.3. Generative adversarial networks based evaluation

This phase focused on generating new data, comprising both normal and attack samples, to further assess the effectiveness of our model. We developed a Generative Adversarial Network (GAN) to produce 1000 normal and 1000 attack samples, which were subsequently used to test the model's performance. Figure 4 displays the loss curves of the generator and discriminator during the generation of normal and attack data, these curves offer insight into the training dynamics and stability of the GAN throughout the learning process. Figure 5 shows a visual comparison between real and generated data samples. Subfigure (a) highlights the similarity between actual and synthetic normal traffic, while subfigure (b) demonstrates how well the GAN replicates attack patterns. These comparisons validate the GAN's ability to produce realistic data distributions for both classes.

When these generated samples were evaluated by the model, it initially achieved an accuracy of 100%. To

Table 8. Model performance with and without GAN-generated data.

Evaluation Setting	Accuracy
Test on Real Data Only	99.94%
Test on GAN Data (avg. over 10 runs)	98.60% ($\pm 1.1\%$)
Combined Real + GAN Testing	99.45%

validate this result and rule out overfitting or model bias toward synthetic data, we conducted multiple evaluation runs using different GAN training seeds and generation conditions. Across 10 independent runs, the model maintained a mean accuracy of 98.6%, with a standard deviation of $\pm 1.1\%$, confirming high robustness and consistency. This variation demonstrates that the model does not overfit to a specific set of synthetic patterns and can generalize effectively across diverse GAN-generated traffic samples.

To further quantify the impact of GAN-generated data on model robustness, we evaluated the model under three distinct settings: testing on real data only, testing solely on GAN-generated synthetic data, and testing on a combined dataset comprising both real and synthetic samples. The results are summarized in Table 8.

4.4. XAI based analysis of GAN generated data

To ensure transparency and interpretability of the model's decisions, we applied the XAI techniques, specifically SHAP and LIME, to the GAN generated data.

The SHAP analysis provides insights into the importance of individual features for the model's predictions on the GAN generated data. Figure 6 presents a summary

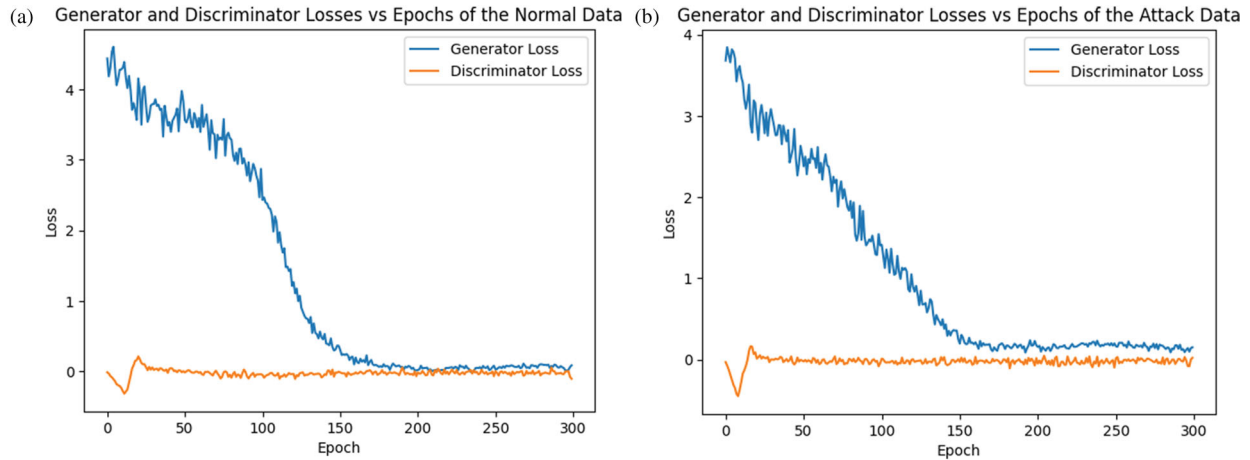


Figure 4. Training loss curves for the generator and discriminator during GAN training. (a) Loss evolution during generation of normal data. (b) Loss evolution during generation of attack data.

plot showing the mean SHAP values for both normal and intrusion predictions.

From the results, the feature **Des_bytes** exhibited the highest importance, indicating a strong impact on the model's decision-making process for both intrusion and normal instances. Other features such as **read_write_physical.process** and **is_syn_only** also contributed significantly, with variations between normal (red) and intrusion (blue) classifications. This reveals the model's sensitivity to these features, especially in distinguishing between attack and normal traffic in the GAN generated data.

In addition to SHAP, we employed LIME to further elucidate the model's decision-making process on the GAN generated data. LIME provides local explanations by approximating the model's complex behavior with simpler, interpretable models for specific instances. Figure 7 showcases the LIME results for a sample classified as normal with a probability of 87% and as an intrusion with a probability of 13%.

The LIME results reveal that the feature **read_write_physical.process** strongly supports the model's prediction of the instance as normal, with a contribution weight of 1.00. In contrast, several features, such as **Des_bytes**, **is_syn_only**, and **Scr_bytes**, influence the model's alternative classification as an intrusion, but with less significant weights.

In this instance, **Des_bytes** was a key feature with a moderate contribution to the potential classification of intrusion, whereas **Duration** and **Scr_bytes** contributed less. The opposing effects of these features underscore the model's capacity to use diverse data characteristics when predicting whether traffic is normal or intrusive. This LIME-based analysis provides a detailed, instance-level interpretation of how the model reacts to GAN generated

Table 9. Summary of SHAP and LIME feature importance across 100 GAN-generated samples.

Feature	SHAP Importance Rank (avg)	LIME Contribution Frequency (%)
Des_bytes	1.0	94%
read_write_physical.process	2.2	87%
is_syn_only	3.1	81%
Scr_bytes	3.7	79%
Duration	4.4	73%

data, offering valuable insights into the model's decision boundaries.

To provide a more comprehensive view of the model's interpretability, we analyzed SHAP and LIME results across multiple GAN-generated instances in the test set. This broader analysis revealed consistent feature influence patterns, strengthening confidence in the model's transparency and reliability.

Table 9 summarizes the frequency and impact of top contributing features across 100 randomly selected GAN-generated samples using both SHAP and LIME explanations.

The SHAP summary plot consistently ranked **Des_bytes** as the most influential feature for both normal and intrusion predictions. This confirms its global importance in shaping model behavior. Similarly, LIME's local explanations frequently highlighted **read_write_physical.process** and **is_syn_only** as decisive factors, especially in cases where the model confidently predicted normal behavior.

These patterns support the conclusion that the BW-LSTM model relies on meaningful, interpretable signals even when tested on GAN-generated data. The synergy between the selected features and their interpretability across both global (SHAP) and local (LIME) explanations

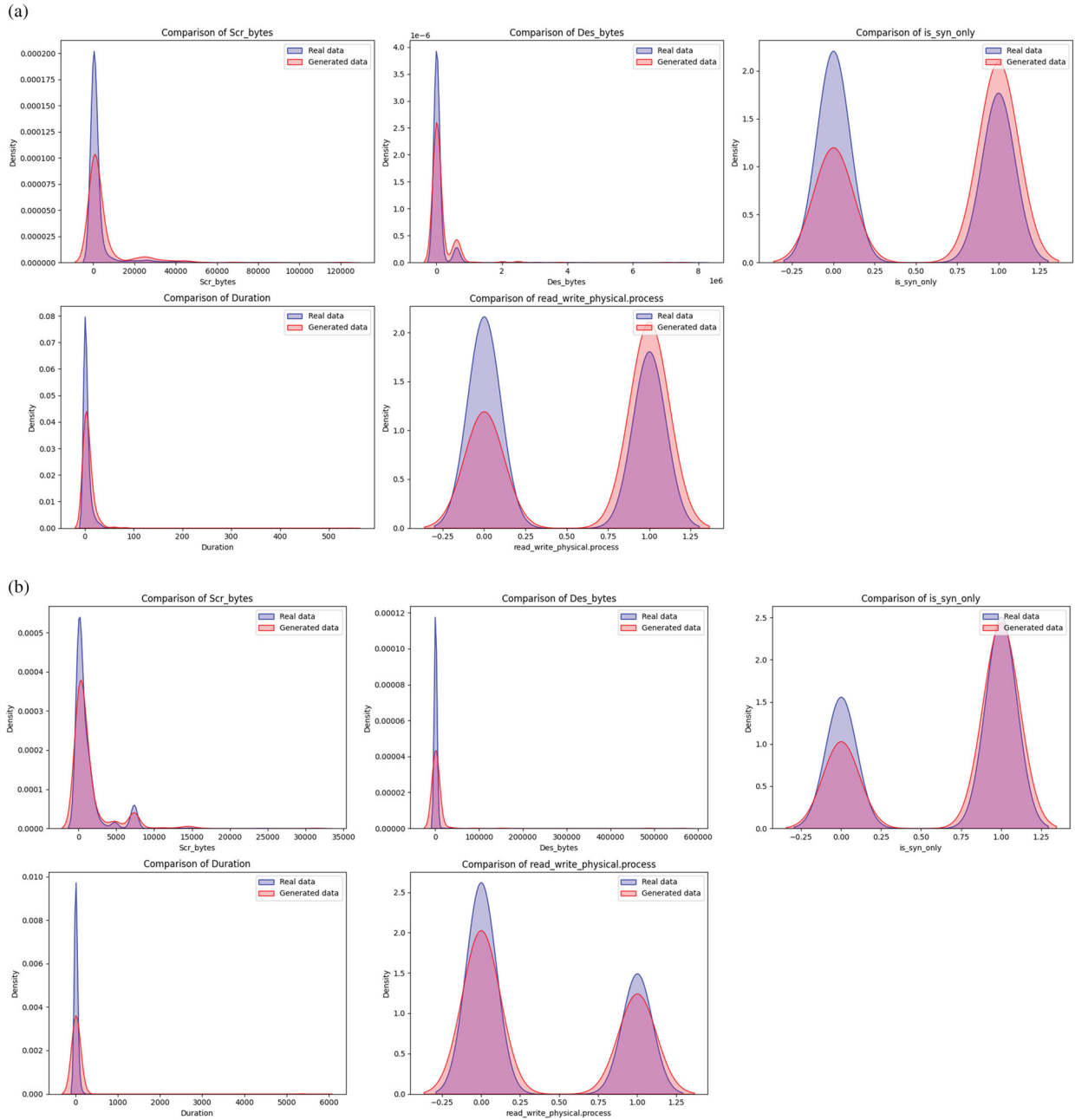


Figure 5. Visual comparison of real and generated samples. (a) Real vs. generated normal data distribution. (b) Real vs. generated attack data distribution.

enhances model trustworthiness a crucial factor for deployment in IIoT environments where human oversight and decision transparency are essential.

5. Result discussion and comparison

Our main objective was to develop an intrusion detection model for the Industrial Internet of Things (IIoT) by classifying network traffic into normal or attack categories. This was achieved using our methodology, which involved feature selection through the H-AGP approach

and model training using our optimized BW-LSTM. Additionally, we incorporated Generative Adversarial Networks (GANs) to generate synthetic data for further evaluation of the model's robustness. To ensure transparency and interpretability, we applied XAI techniques, specifically SHAP and LIME, on the trained model with GAN generated data to analyze the model's decision-making process. The interpretation of the obtained results is presented in this section.

In our preprocessing stage, we applied a correlation threshold of 0.7 to eliminate highly correlated features,

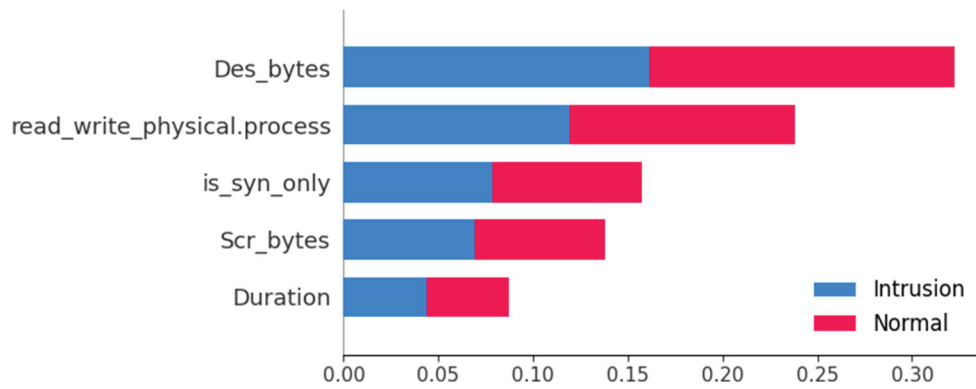


Figure 6. SHAP summary feature importance for the GAN generated data.

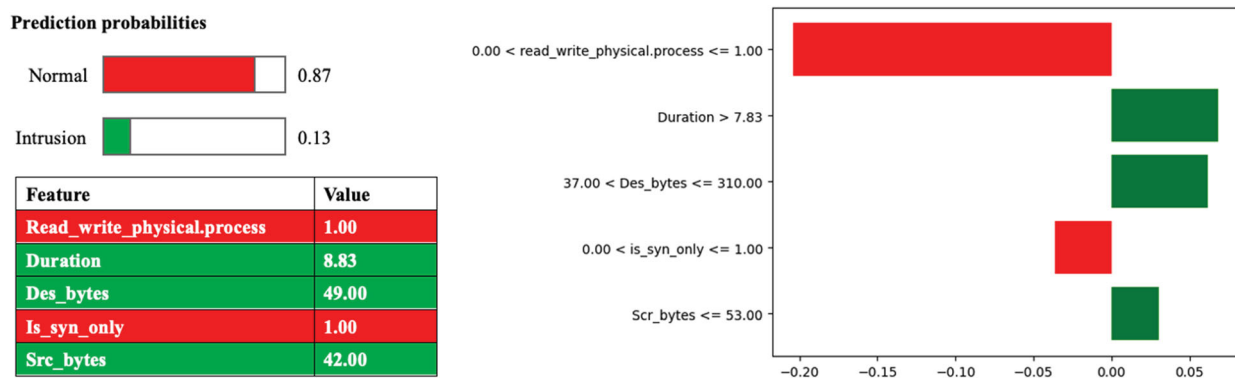


Figure 7. LIME explanation for the first instance in the test set using GAN generated data.

ultimately reducing our feature set to 32. This step is critical as it helps to mitigate multicollinearity, which can distort model predictions by making the model overly sensitive to small changes in the input data. By removing these redundancies, we ensured that each feature contributes unique information, enhancing the model's interpretability and performance.

In the feature selection stage, we employed the H-AGP approach for feature selection. Our approach identified five features which are: duration, des_bytes, src_bytes, is_syn_only, and read_write_physical.process, yielding an impressive accuracy of 99.5%. These features provide a mix of network-level indicators and physical process-level indicators, which can offer a more comprehensive view of both network and physical anomalies. The parameters we chose played a big role in achieving this accuracy in feature selection.

As result of the feature selection stage, this approach provides a balanced set of features that cover both network behavior and physical process activities, potentially leading to a more robust intrusion detection system.

The high accuracy achieved with only five selected features can be attributed to the synergistic nature of the H-AGP approach, which combines Ant Colony Optimization (ACO), Genetic Algorithm (GA), and Particle Swarm Optimization (PSO). ACO contributes by

prioritizing paths that historically lead to better performance using pheromone trails, thus reinforcing consistently relevant features. GA introduces population diversity through crossover and mutation, which helps escape local optima and explore new combinations of features. PSO accelerates convergence toward optimal feature subsets by adjusting candidates based on both personal and global best solutions. This hybridization ensures a more comprehensive search of the solution space, effectively balancing exploration and exploitation. As a result, H-AGP identifies a compact yet highly discriminative subset of features that retain critical information for classification, leading to superior accuracy with minimal input dimensionality. Furthermore, by reducing feature redundancy and noise, the model becomes more generalizable and less prone to overfitting, which is essential in IIoT environments where data diversity is high and false positives can be costly.

The results from the BW-LSTM approach are highly impressive, demonstrating exceptional performance in the training and evaluation phases. Each parameter and decision in our methodology was carefully justified to enhance model performance.

The evaluation metrics (precision, recall, f1-score and accuracy) underscore the robustness of our model. High precision indicates a very low false positive rate, while

Table 10. Comparison of different intrusion detection approaches.

Ref	Dataset	Feature selection techniques	Features	Training techniques	Accuracy
[11]	X-IIoTID	?	59	Residual Network (ResNet), Federated learning approach	99.16%
[13]	NSL-KDD, UNSW-NB15, X-IIoTID	Extreme random forest algorithm	?	Dilated Convolution, Dual Attention Mechanism	99.68%, 98.50%, 99.85%
[17]	Edge-IIoT	Feature Importance	8	Lightweight Stacking Ensemble Learning	87.37%
[21]	CICIDS2017, X-IIoTID	Self-adaptive attention mechanism	?	Bidirectional Long-Short Term Memory (BiLSTM), Self-adaptive attention mechanism	99.92%, 96.54%
[14]	Edge-IIoTSet, X-IIoTID	Extra-tree classifier	?	Self-attention layer, Convolutional neural networks (CNN) layers	99.66%, 99.72%
[15]	Edge-IIoTSet, WUSTL-IIoT-2021, X-IIoTID	Autoencoder, Modified locally linear embedding	?	Convolutional Neural Network (CNN), Gated Recurrent Unit (GRU), Bayesian optimization	97.68% (with combined dataset), 97.80% (when trained in X-IIoTID and tested on Edge-IIoTSet)
[16]	IIoT2D20, Edge-IIoT	?	56	Self-Attention-Based Deep Convolutional Neural Network (SA-DCNN)	96.89%, 99.95%
Our work	X-IIoTID	H-AGP Algorithm (Hybrid ACO – GA – PSO)	5	BW-LSTM (Optimized LSTM with Bat Algorithm and Whale Optimization Algorithm)	99.94%

high recall shows the model effectively identifies true positives. The F1-Score, balancing precision and recall, along with the accuracy confirm the model's overall excellence. These metrics are particularly crucial in applications where both false positives and false negatives carry significant consequences.

While the initial evaluation of GAN-generated samples yielded a perfect accuracy score, further analysis was conducted to ensure this was not an artifact of overfitting or low diversity in synthetic data. By varying the GAN's initialization and training conditions over multiple runs, we observed a consistently high performance, with only minimal variance. This strongly suggests that the GAN was able to generate realistic and representative samples, closely approximating the distribution of the original IIoT dataset. These results reinforce the model's ability to generalize to novel inputs, even when derived from artificially synthesized data.

The SHAP and LIME analyses validate the interpretability and robustness of our BW-LSTM model in classifying IIoT network traffic, even with GAN generated data. The features identified through H-AGP feature selection played a significant role in the model's decisions. This approach enhances both performance and transparency, critical for effective and secure IIoT systems.

As a simple comparison with previous works, our work outperforms Works [11,16] in selecting the least number of features. Also, our model outperforms Works [11,13,14,21] in achieving the highest accuracy when using the same X-IIoTID dataset. In contrast to the works [19,20], which focus primarily on using GANs for addressing dataset imbalance in intrusion detection, our approach incorporates GANs not only for generating

synthetic data but also for evaluating the robustness of the model on unseen data. Additionally, unlike the studies [22,23], which apply XAI methods to deep learning models, we integrate XAI techniques such as SHAP and LIME specifically on models trained with GAN-generated data, enhancing transparency and interpretability for synthetic data evaluations. This demonstrates the power of our approach to build a robust IIoT intrusion detection model starting from data collection and feature selection using H-AGP algorithm to model training and evaluation using BW-LSTM approach. Table 10 summarizes the comparison of our work with previous works in terms of dataset, feature selection techniques, number of features, training techniques and accuracy achieved.

Finally, with regard to practical deployment considerations, it is important to recognize that real-world IIoT environments often operate under resource constraints such as limited processing power, memory, and energy. Our approach is intentionally designed to be deployment friendly. Although the training phase involves computationally intensive processes such as hybrid optimization for feature selection and LSTM model tuning, these are conducted offline. During deployment, only the five features identified by the H-AGP algorithm are extracted from real-time traffic and passed to the pretrained BW-LSTM model for inference. This design significantly reduces the computational overhead on edge devices. By offloading training tasks to centralized or cloud-based systems and deploying a lightweight inference model at the edge, our solution offers both practicality and efficiency for IIoT environments. Future work may explore further enhancements such as model compression, quantization, or edge-specific optimization techniques to further improve inference speed and energy efficiency.

6. Conclusion

The Industrial Internet of Things (IIoT) merges industrial operations with advanced connectivity through sensors, software, and networks, driving both efficiency and safety. However, this interconnectivity exposes IIoT systems to potential security vulnerabilities, making effective intrusion detection essential for maintaining data integrity and system reliability.

In this research, we proposed a hybrid approach to intrusion detection using the X-IIoTID dataset. Our Hybrid ACO-GA/PSO (H-AGP) method combines Ant Colony Optimization (ACO), Genetic Algorithm (GA), and Particle Swarm Optimization (PSO) to select the most critical features, resulting in improved detection performance. Additionally, the BW-LSTM method, utilizing the Bat Algorithm and Whale Optimization Algorithm, optimizes Long Short-Term Memory (LSTM) networks by fine-tuning hyper-parameters, further enhancing classification accuracy.

We successfully met our objective of classifying IIoT network traffic as either normal or malicious. The H-AGP approach effectively selected relevant features, and the BW-LSTM method achieved a high detection accuracy. Our model's robustness was further validated through the integration of Generative Adversarial Networks (GANs) to generate synthetic data for unseen traffic, which the model accurately classified, demonstrating its ability to generalize beyond the training data.

Furthermore, we applied Explainable AI (XAI) techniques, specifically SHAP and LIME, to the model trained on GAN generated data, providing insights into the decision-making process and enhancing model transparency. This helps in understanding which features have the greatest impact on classifying network traffic, thus aiding in trust and reliability in practical deployment.

In conclusion, our hybrid approach offers a powerful and practical solution for improving security in IIoT environments by enhancing intrusion detection accuracy and robustness. Designed with resource constrained IIoT systems in mind, the computationally intensive training is performed offline, while real-time inference relies on only five selected features minimizing the burden on edge devices and ensuring deployment feasibility. Our research contributes to the development of reliable, explainable, and efficient intrusion detection mechanisms, making it well suited for real-world IIoT applications. Future work will explore further optimizations, such as model quantization and edge-specific adaptations, to enhance efficiency and responsiveness in evolving IIoT systems.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Author contributions

CRedit: **ABID Dhiya Eddine**: Conceptualization, Data curation, Formal analysis, Methodology, Software, Visualization, Writing – original draft, Writing – review & editing; **GHAZLI Abdelkader**: Conceptualization, Project administration, Supervision, Validation; **BOUACHE Mourad**: Conceptualization, Methodology, Project administration, Supervision

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